EW Customized Tutor

English, Economics, Politics, and Philosophy

Econometric

Unit 4 Quantitative Methods and its Application

Lesson 4.1 Basic Charts in Excel

Pertinent Topics:

Probability: the chances/potential of an event occuring

- Statistics vs Probability

Statistics: empirical (experience based) data (numbers)

- Bar Charts, Histogram, Line Charts, Scatter Plots, Pie Chart

Success Criteria:

- I will learn about the 'parameters' in quantitative method.
- I will learn about the types of Charts in Excel
- I will be able to visually represent a set of data in Excel

MINDS ON

In our life we encounter various types of charts representing some statistics. But how do we determine which type of chart to be used? Furthermore, what is the ‘rationale’ behind the representation of each chart? What does terminology such as mean, median, correlation mean? Do you think data is always an accurate portrays of an event? Bias: inaccurate representation

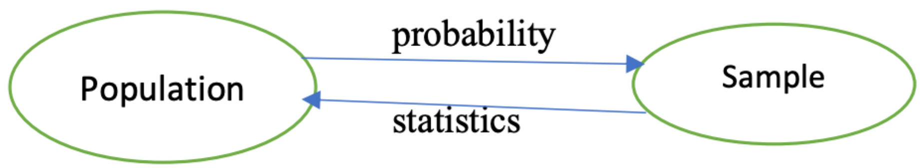
Q: what if we have a lot of people, how do collect the statistics that represent these people?

You collect data from a smaller group of people and generalize to represent all the people

1. Understand the population
2. Sample people based on the understanding
3. Apply probability theory to ‘approximate’ the population from our sample

I. Statistics vs Probability

A. Before we venture into creating our own charts. Let’s get familiar with some terminologies that may seem important later.



Parameters

Sample statistics
(Point estimators, Estimators)

Parameters: values/indicator

$$\text{Statistics}\begin{cases}\text{Descriptive Statistics}\\ \\ \text{Inferential Statistics}\begin{cases}\text{Estimation}\begin{cases}\text{Point estimstion}\\ \text{Interval estimation}\end{cases}\\ \text{Hypothesis Testing}\end{cases}\end{cases}$$

Broadly speaking, we collect data from a *sample* that represents a larger *population*. We then, by quantitative method, and theories in probability, approximate/measure the *parameters* in the large population. The reason for this to be a seemly complicated yet effective way of conducting a statistical analysis is because often in real life it is rather unrealistic for us to collect data for an entire population. For example, if we were to know the voting trend before an election, it seems unrealistic and unfeasible for us to sample all population in Taiwan (>15million people).

B. Parameters: numbers/values

Parameter name	Population parameter symbol	Sample statistic
Number of cases	N	n
Mean Average	μ (mu)	$\bar{x}$ (Sample mean)
probability Proportion	π (Pi)	P (Sample proportion)
Variance	σ^2 (Sigma-square)	s^2 (Sample variance)
Standard deviation	σ (Sigma)	s (sample standard deviation)
Correlation	ρ (rho)	r (Sample correlation)
Regression Coefficient	β (beta)	b (sample regression coefficient)

1. Number of Cases (N/n): Represents the total number of observations in a population (N) or sample (n). Example: If surveying 100 students in a school, the sample size (n) is 100.
2. Mean ($\mu/\bar{x}$): The average value of a dataset, calculated as the sum of all values divided by the number of observations. Example: The mean score of a class of students in a math test.

3. Proportion (π/P): The fraction of the population or sample that exhibits a specific characteristic. Example: If 30 out of 100 students prefer online classes, the proportion is 30%.
4. Variance (σ^2/s^2): Measures how data points differ from the mean, indicating the spread of data. Example: The variance in test scores among students in a class.
5. Standard Deviation (σ/s): The square root of variance, providing a measure of dispersion in the same units as the data. Example: The standard deviation of daily temperatures in a week.
6. Correlation (ρ/r): A statistic that shows the strength and direction of a relationship between two variables. Example: The correlation between hours studied and test scores.
7. Regression Coefficient (β/b): Indicates the change in the dependent variable for a unit change in the independent variable. Example: In a study on the impact of exercise hours on weight loss, β represents the amount of weight loss per hour of exercise.

For now, we rely on Excel function to calculate the ‘value’ of these parameters.

Concerning the ‘comparison’

II. Charts

1. Bar Charts

something in words

Explanation: Displays categorical data using rectangular bars, where the length of each bar represents the value of a category.

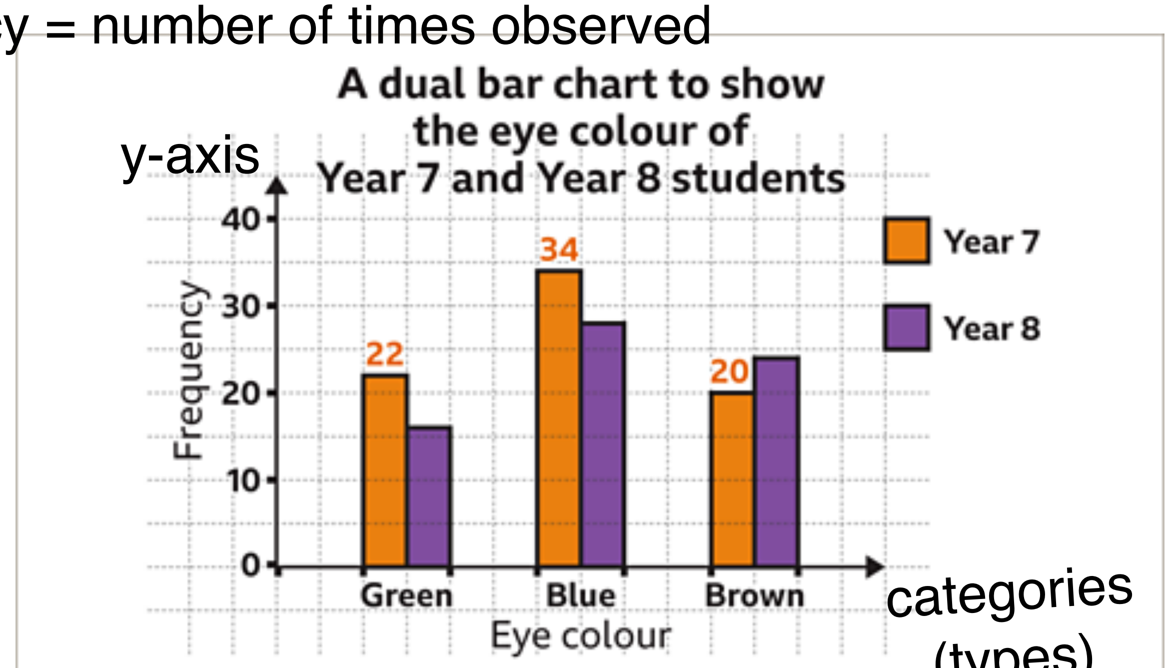
Contextual Usage: Used to compare quantities across different categories, such as sales by product or population by region.

Rationale: Makes it easy to compare discrete groups visually and

categorical

identify the highest or lowest values.

Frequency = number of times observed



categories (types) x-axis

Observation: In year 7, 22 people have green eyes, 34 have blue eyes, and 20 have brown eyes; in year 8, 16 people have green eyes, 28 people have blue eyes, and 24 people have brown eyes.

alternative intepretation:

22 people in year 7 and 16 people in year 8 have green eyes …

Concerning the ‘distrubution’

2. Histogram

quantitative

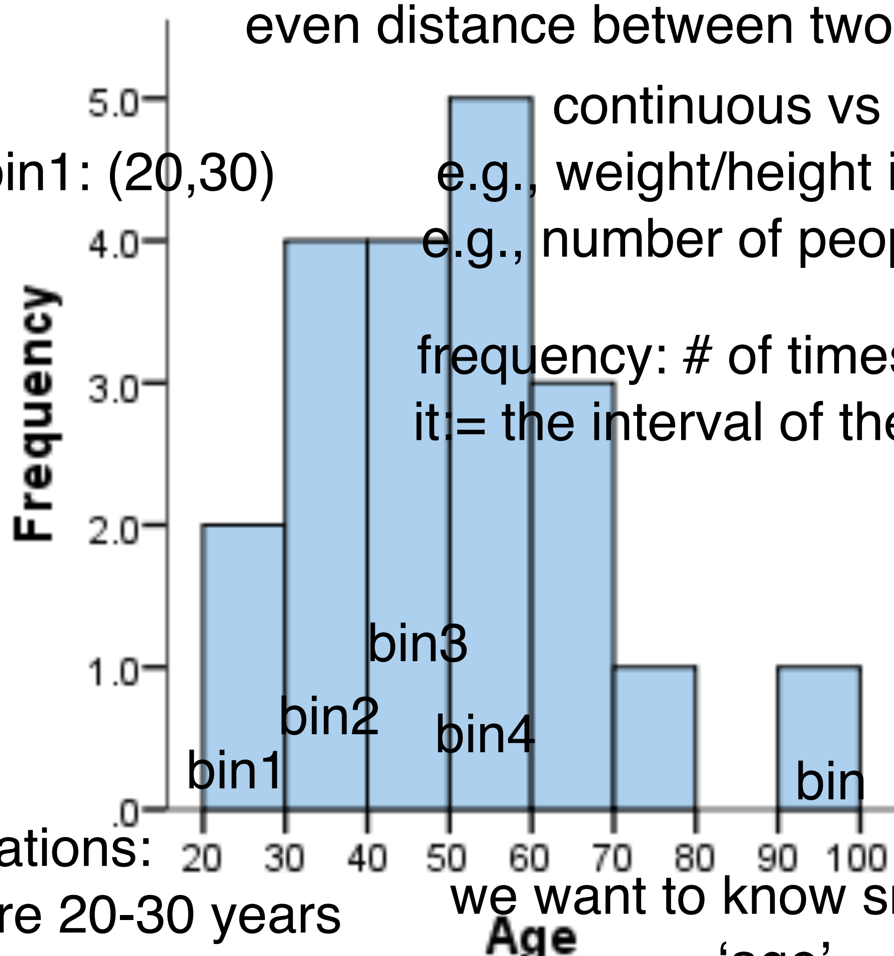
Explanation: Represents the distribution of numerical data using contiguous bars, grouped into intervals (bins). Contextual Usage: Commonly used to visualize the frequency of data within certain ranges, like age distributions or exam scores. Rationale: Helps identify patterns such as skewness, spread, or central tendencies in continuous data.

bin: space or gap between two things
even distance between two things

continuous vs discrete
e.g., weight/height is continuous
e.g., number of people is discrete

bin1: (20,30)

frequency: # of times it occurs
it:= the interval of the category



Observations:
2 people who are 20-30 years old, 4 people for 30-40, 4 in (40,50), 5 in (50,60), 3 in (60,70), 1 in (70,80), 0 in (80,90), 1 in (90, 100).

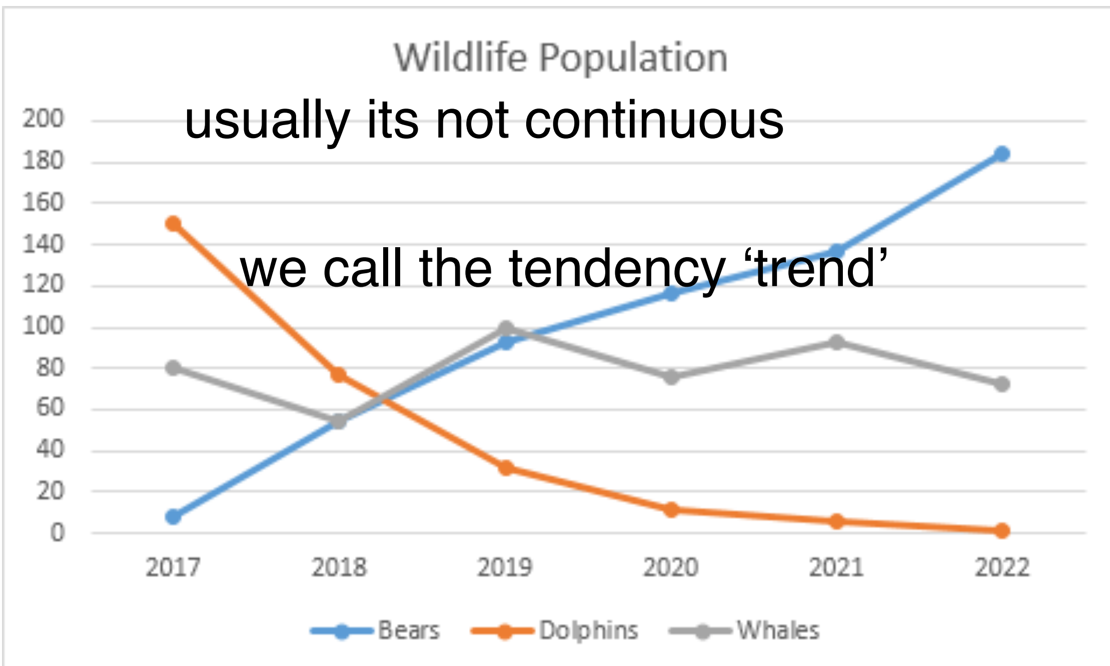
we want to know smt about ‘age’
in particular, the distribution of age
0 - 100 is typically possible for a given population

Concerning the ‘trend’

3. Line Charts

Explanation: Uses points connected by lines to show trends over time or sequential data. Contextual Usage: Ideal for showing changes over time, such as stock prices, weather data, or website traffic.

Rationale: Effective in highlighting trends, fluctuations, and patterns over a continuous range.



usually its not continuous

we call the tendency ‘trend’

Observation:

(1) Dolphins’ population is declining. Whilst its population was large in 2017, it has no longer been the case in 2022.

(2) Bear’s population have been increasing and peak at 2022.

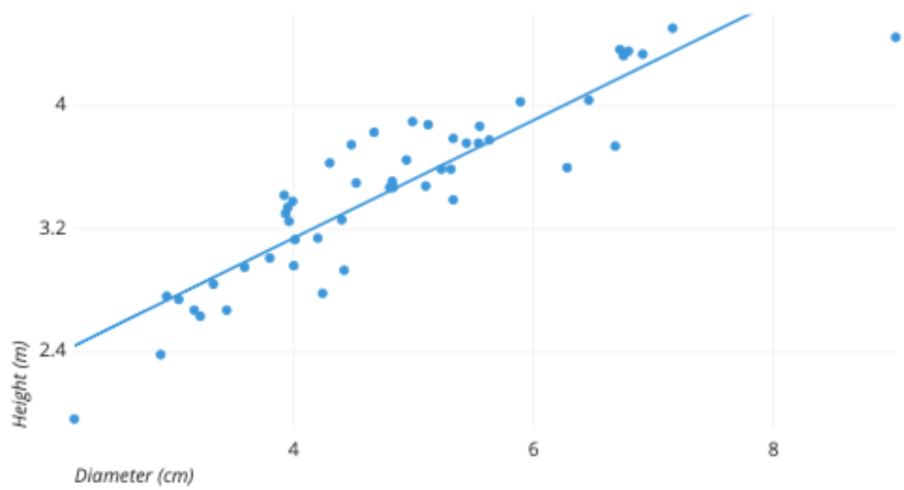
(3) Whales’ population fluctuates over time and seem to stabilize at 80.

Bears population surpasses Dolphins’ in 2019 (not in 2017)

Concerning the ‘Correlation’

Scatter Plots SUPER IMPORTANT

Explanation: Plots points on a grid to show the relationship between two variables. Contextual Usage: Useful for identifying correlations or trends, such as the relationship between study hours and grades. Rationale: Enables visualization of the strength, direction, and nature of relationships between variables.

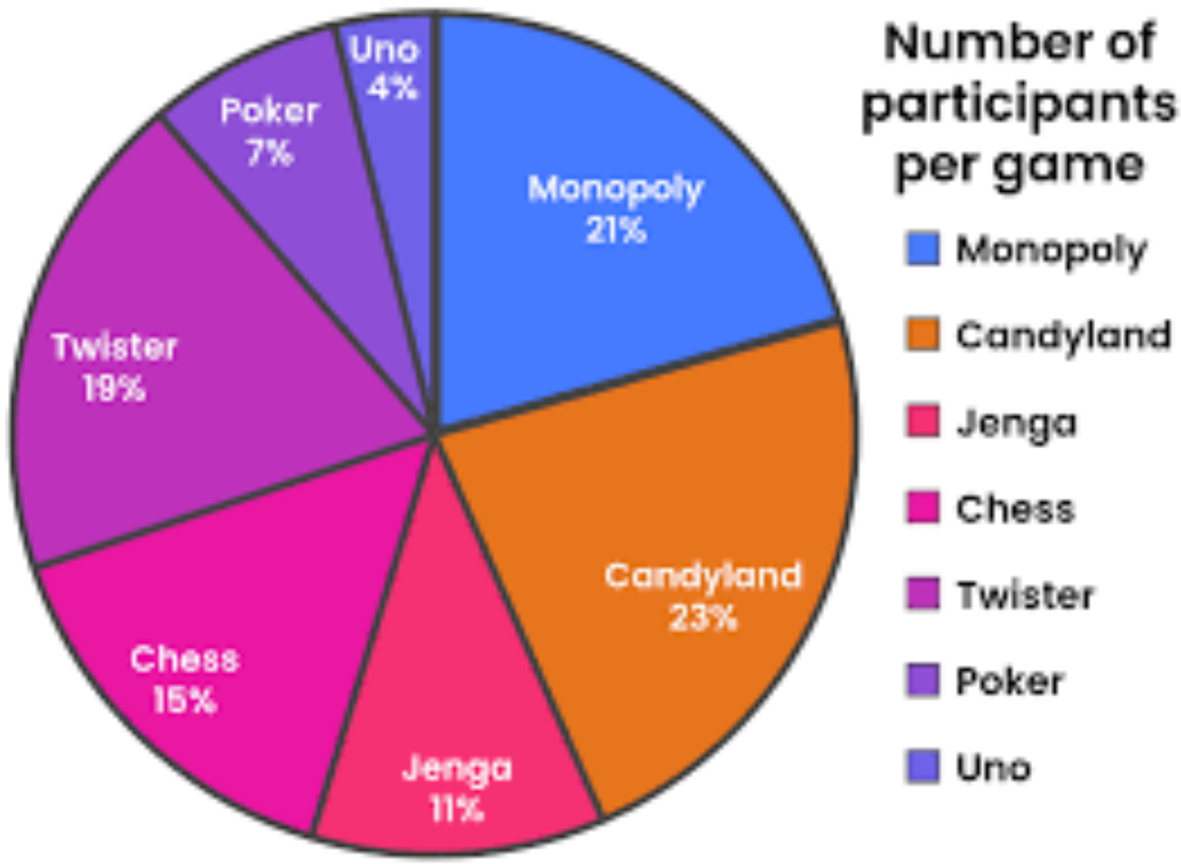


Pie Charts

Divides a circle into slices to represent proportions of a whole.

Contextual Usage: Commonly used for showing percentages or parts of a total, such as market share or budget allocation.

Rationale: Intuitively displays how parts contribute to a whole, though less effective for precise comparisons.



Some techniques in Excel:

Suppose we are dealing with a large set of data.

How can we utilize excel to extract the desired information?

How do we mitigate statistics bias? Ans: increase sample size